

Industrial Employee Safety and Occupational Health Management System

DSA0101-OOPSwitC++ / SlotD / CapstoneProject-Review2 / Team8

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Abstract

- Industrial workplaces such as factories, warehouses, and construction sites face constant risks of accidents, injuries, and health hazards from unsafe conditions, hazardous gases, and machinery.
- Manual safety inspection and paper-based incident reporting are slow, error-prone, and reactive rather than preventive.
- This project proposes an Industrial Employee Safety and Occupational Health Management System that digitizes safety compliance, tracks employee health records, monitors hazard zones, and automates incident reporting and alerts.
- The system is built using Object-Oriented Programming concepts in C++ to model employees, safety equipment, hazard zones, and incident records as reusable, maintainable classes.
- The goal is to reduce workplace accidents, ensure regulatory compliance, and improve response time during emergencies.

Existing System

- Safety inspections and incident reports are largely manual and paper-based, causing delays in identifying hazards.
- No centralized system tracks employee health history, PPE compliance, and training records together.
- Hazardous zones (gas leaks, high temperature, restricted machinery areas) are not monitored in real time.
- Emergency alerts depend on manual reporting, increasing response time during accidents.
- Existing IoT-based solutions from literature focus mainly on sensor hardware and lack an integrated, extensible software backend.
- There is a lack of role-based access for supervisors, safety officers, and employees within a single management system.

Problem Statement

Manual, paper-based safety inspection and incident reporting delay hazard detection and corrective action.

Employee health, PPE compliance, and training records are fragmented across different registers, making them hard to track together.

Hazardous zones are not monitored continuously, so unsafe conditions can go unnoticed until an accident occurs. Emergency response is slow because alerts rely on someone manually noticing and reporting a hazard.

There is no unified, software-driven system to manage employees, hazard zones, incidents, and compliance under one platform.

Objectives

- Develop an Industrial Employee Safety and Occupational Health Management System using C++.
 - Apply Object-Oriented Programming concepts to model employees, hazard zones, and incidents as reusable classes. Store and manage employee profiles, health records, and training/certification status.
 - Classify hazard zones by risk level and log sensor-based hazard events.
 - Digitize incident and near-miss reporting with severity classification and corrective action tracking.
 - Track PPE issuance and compliance status for every employee.
 - Generate real-time alerts and notifications to supervisors during hazard or incident events.
- Improve workplace safety, regulatory compliance, and emergency response time.

Literature Survey (Latest Papers)

Year	Title / Authors	Key Contribution
2024–25	Smart IoT System for Real-Time Worker Safety Monitoring in Dead Zones (IEEE, 2025)	Wearable sensors with ML-based fall detection and location tracking in connectivity-limited zones.
2024–25	Comprehensive IoT-Based Monitoring System for Worker's Health and Safety in High-Risk Environments (IEEE, 2024)	Real-time SpO2, heart rate, gas and temperature monitoring with dashboard alerts.
2023–24	Shah, I.A. & Mishra, S. — AI in Advancing Occupational Health and Safety (2024)	Reviews how AI/ML predicts and prevents workplace hazards, improving OHS decisions.
2023–24	IoT-Enhanced Workplace Safety for Real-Time Monitoring and Hazard Detection (IEEE, 2024)	Sensor network architecture for environmental and wearable monitoring with hazard analytics.
2022–23	IoT Based Wearable Device for Workers in Industrial Scenarios (IEEE, 2023)	ESP-32 based wearable tracking gas exposure, pulse rate, body temperature and emergency alerts.
2022–23	IoT based Smart Bracelet for Worker Safety and Health Monitoring (IWAIIP, IEEE, 2023)	Accelerometer-based vibration monitoring for machine operators with mobile/web notifications.

Proposed System / Solution

Employee & Health Management Module

Stores employee profiles, health records, and training/certification status, using OOP principles such as classes, inheritance, and encapsulation.

HazardZone & Alert Module

Models zones with risk levels, links sensor data to affected employees, and automatically raises real-time alerts when thresholds are breached.

Incident & Compliance Module

Digitizes incident / near-miss reporting with severity classification and tracks PPE issuance and compliance status per employee.

System Modules

Employee Management Module

Add, update, and view employee details, department, and role.

Health & Medical Records Module

Maintains periodic health checkup data and flags at-risk employees.

Hazard Zone Monitoring Module

Classifies zones by risk level and logs sensor-based hazard events.

Incident & Accident Reporting Module

Records incidents with cause, severity, and corrective action taken.

PPE & Compliance Tracking Module

Tracks safety equipment issued and compliance verification.

Alert & Notification Module

Sends real-time alerts to supervisors during hazard or incident events.

Module 1: Employee & Health Management

OOP Concepts Used	Functions	Output
• Classes and Objects • Encapsulation • Constructors • Inheritance	• Add new employee records • Update employee & department details • Store periodic health checkup data • Flag at-risk employees • Track training / certification status • Search employee by ID or name	• Updated employee database • Accurate health & training records • List of at-risk employees

Module 2: Hazard Zone Monitoring & Alerts

OOP Concepts Used

- Inheritance
- Polymorphism
- Virtual Functions

Functions

- Classify zones by risk level
- Log sensor-based hazard events (gas, temperature)
- Link hazard data to affected employees
- Auto-raise alert when threshold is breached
- Notify supervisors in real time

Output

- Real-time hazard alerts
- Zone risk classification log
- Faster emergency response

Module 3: Incident, PPE & Compliance

OOP Concepts Used

- Encapsulation
- File Handling
- Abstraction

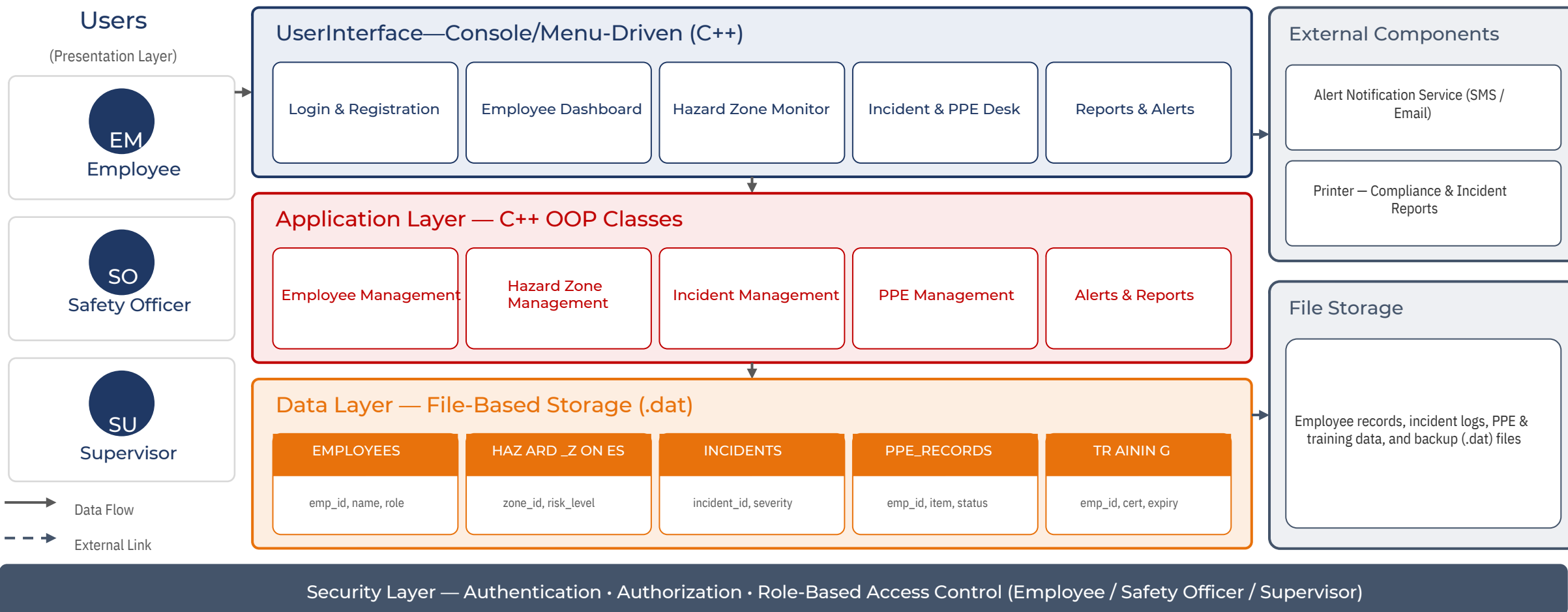
Functions

- Record incidents with cause & severity
- Log corrective action taken
- Track PPE issuance per employee
- Verify PPE compliance status
- Generate compliance & incident reports

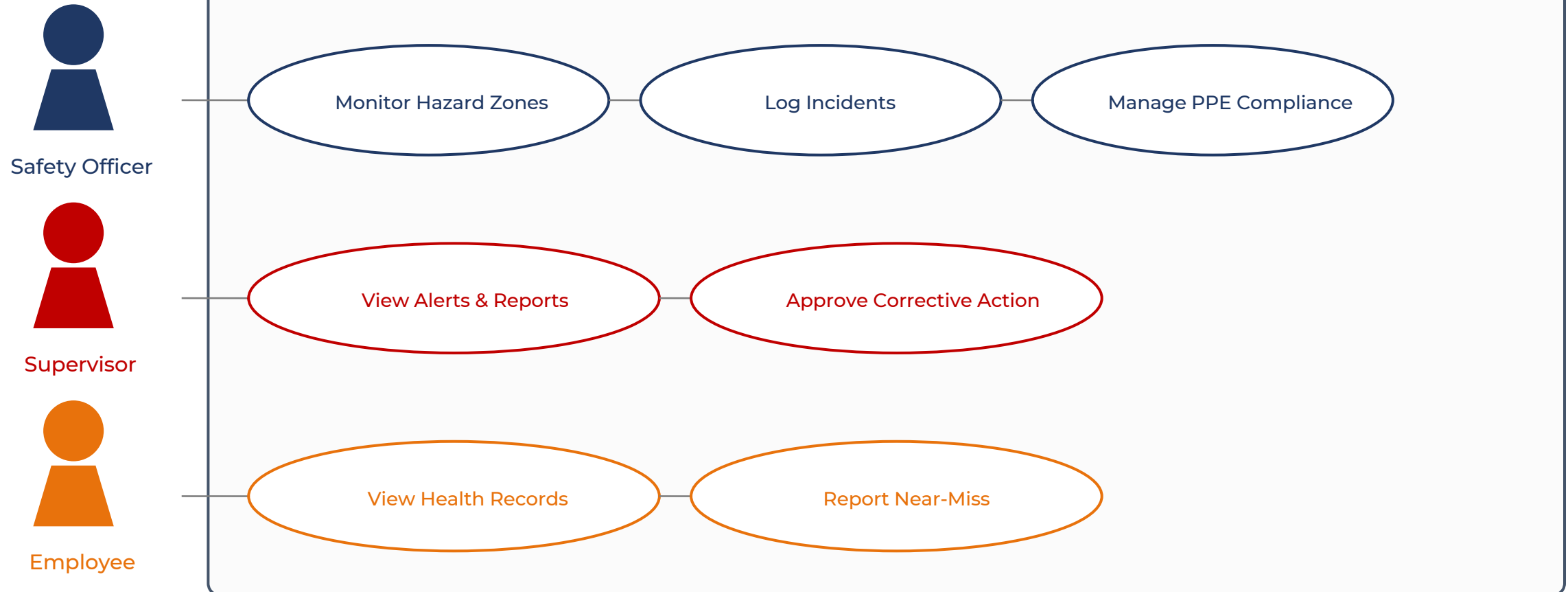
Output

- Digitized incident records
- PPE compliance report
- Regulatory audit trail

System Architecture



Use Case Diagram



System Workflow



Continuous loop: hazard zones and PPE compliance are re-checked after every logged event.

Tools and Technologies

Software

- Code::Blocks / Dev-C++
- GCC Compiler
- Windows Operating System

Technologies Used

- C++ Programming Language
- Object-Oriented Design
- File Handling for persistence

OOP Concepts

- Classes & Objects
- Encapsulation
- Inheritance
- Polymorphism
- Abstraction

OOP Concepts Used

- Classes and Objects – represent employees, hazard zones, incidents, and PPE items as system entities.
- Encapsulation – combines employee, health, and compliance data with the functions that operate on them.
- Inheritance – supports reusable class structures across employee roles and hazard-zone types.
- Polymorphism – provides different alert / reporting behaviours through function overriding.
- Abstraction – represents essential safety operations while hiding internal implementation detail.
- Constructors – used to initialize employee, zone, and incident objects with default/starting data.
- File Handling – used for storing and retrieving employee, incident, and compliance records.

Advantages

- Replaces manual, paper-based safety processes with a unified digital system.
- Provides real-time hazard-zone monitoring and faster emergency alerts.
- Centralizes employee health, PPE compliance, and training records in one place.
- Reduces human error in incident reporting and safety documentation.
- Improves regulatory compliance and audit readiness.
- Modular OOP-based design makes the system easy to maintain and extend.
- Supports faster, more informed decision-making by supervisors and safety officers.

Future Scope

- Integration with real-time IoT sensors for gas, temperature, and wearable health monitoring.
- Machine learning-based predictive analytics to anticipate hazards before they occur.
- Mobile application for supervisors and employees to receive alerts remotely.
- Cloud-based storage for centralized, multi-site safety data management.
- Dashboard-based visual analytics for hazard trends and compliance tracking.
- GPS / location-based tracking of employees in large industrial sites.

Conclusion

- The proposed Industrial Employee Safety and Occupational Health Management System replaces manual, fragmented safety processes with a unified, object-oriented software solution.
- By integrating employee records, hazard monitoring, incident reporting, and PPE compliance into one system, organizations can respond faster to emergencies and reduce workplace accidents.
- The modular OOP-based design allows the system to be extended in future with IoT sensor integration and predictive analytics for proactive hazard prevention.
- Overall, the system contributes to a safer, more compliant, and more accountable industrial workplace.

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THANK YOU

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